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Design engineer building real-time graphics and generative-media interfaces.

University of Pennsylvania, BSE Computer Graphics + MSE Computer Science c/o 2027

Currently : GPU Engineering + Systems coursework (2026–27) **Exploring:** generative-media tools & inference pipelines.

PROJECTS

digital loom — GenAI, Three.js, GLSL, TypeScript **2026**

Turn a photo of fabric into cloth you can drape in 3D! AI-extracted PBR materials, additional derived shader maps, parallax occlusion mapping + edge fraying for realism, and a custom warp/weft solver for the physics.

Internet Atlas — Three.js, D3.js, React, Pinecone, FastAPI, Supabase **2025**

An ML embedding pipeline places websites by visual and textual feel; browsing data becomes a navigable 3D galaxy (300 nodes, 1000+ edges). Technical lead: built the 3D graph and all interactions, designed the data model, connected the backend, demo'd live to 40 simultaneous users on a \$5 hosting budget & made the deployment static afterwards.

Mini Minecraft — C++, OpenGL, GLSL, Qt **2024**

Voxel engine from scratch in C++/OpenGL: multi-threading and chunking, procedural terrain, sky shaders, real-time water waves and reflections (team of 3).

Into the Blue: Virtual Experience for Penn Archaeology Museum — React, Next.js **2024**

Offline-first artifact hunt. Snap an artifact, keep it as a sticker, all in the browser. Built camera to sticker pipeline (multi-path outline tracing, screenshot-time clipping, etc.) and IndexedDB storage. 1 year, 180k visitors (team of 5).

Simulation & Generative Experiments (live at estellekimdev.com/playground) **2025~**

WORK EXPERIENCE

SWE Intern @ PayPal — San Jose, CA **2026**

- Building CTO-level initiative to surface hidden friction in the checkout flow. Digital Wallet platform team.
- Designed multi-dimensional data visualization & hypothesis testing tool, working with senior engineers and PMs.

Mobile App UI Designer @ tally — New York, NY | Remote **2026**

- Designed and specified end-to-end mobile UI, interaction flows, and state logic for a habit-based fintech app.
- Solo designer: built interaction patterns and interface states spanning contracts, onboarding, and verification.

Data Visualization & Engineering Intern @ Southern California Edison — Rosemead, CA **2025**

- Designed and optimized ETL pipelines serving a utility with 15M residents, delivered solutions to 10 + internal teams

Frontend Web Developer @ Children's Hospital of Philadelphia — Philadelphia, PA **2025-6**

- Contributed to ML-driven research management hub for autism studies using React, TypeScript, and REST APIs.

ACTIVITIES

Vice President (External), 3x Project Lead, Full-Stack Developer @ Penn Spark **2024~**

- Lead corporate partnerships and recruiting / career events for Penn's community of creative devs and designers.
- Led redesign and rebuild of Spark's website, improving information architecture, branding, and maintainability.

Industrials Committee @ Wharton Investment & Trading Group **2024-6**

- Applied financial modeling + competitive analysis, and pitched in intercollegiate competitions.

Governmental Affairs Director @ California Association of Student Councils **2024**

- Authored & passed 3 CA laws, facilitated a 20-organization coalition including ACLU and Human Rights Watch.

SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, GLSL/WGSL, HTML/CSS

Frameworks: React, Three.js, OpenGL, WebGL/GPU, Next.js, React Native, Node.js, Django, PostgreSQL, Supabase

AI / Gen-Media: PyTorch, LLM transformers, Computer Vision, Image/Video/3D genAI, Claude Code / Codex

Tools: Figma, Adobe Creative Suite, Maya, Docker, AWS, Github, Git, Arduino (Electronics), Blender, TouchDesigner

Achievements: 3× AIME qualifier (top 2.5% nationwide); Korean TOPIK II Level 6 (highest).